

Compression-based Joint Channel Estimation and Equalization in Low-Altitude MIMO System

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Abstract—Unmanned aerial vehicle (UAV)-assisted communications can extend uplink coverage for low-altitude economy (LAE) services by using a UAV as an aerial remote radio unit and offloading baseband processing to a ground baseband unit (GBU). However, limited fronthaul capacity and stringent latency constraints make it difficult to support computationally intensive baseband processing, such as channel estimation and equalization, especially in multi-carrier MIMO systems. To address these challenges, we propose a linear compression-based joint channel estimation and equalization (LC-CE) scheme, where the UAV employs a shared compression matrix to reduce the dimensionality of received signals across all subcarriers. The resulting optimization problem is challenging due to the coupling between the shared compression matrix and the subcarrier-wise equalization matrices. Leveraging the Grassmann manifold structure, we develop an efficient Riemannian gradient descent (RGD) algorithm and a low-complexity heuristic alternative for practical deployment. Simulation results demonstrate that the proposed LC-CE scheme significantly reduces fronthaul overhead while maintaining performance close to the uncompressed baseline.

Index Terms—UAV, multi-carrier, MIMO, channel estimation, equalization, compression

I. INTRODUCTION

The low-altitude economy (LAE) is fostering uplink-intensive applications such as high-definition video streaming, smart sensing, and emergency response [1, 2]. In these scenarios, unmanned aerial vehicles (UAVs) can be deployed as aerial communication platforms that extend coverage to areas poorly served by terrestrial infrastructure, such as areas with severe blockage and temporary hotspots. In a practical UAV-assisted architecture, the UAV serves as an aerial remote radio unit (RRU) that collects uplink signals, while most baseband processing is offloaded to a ground baseband unit (GBU) due to the UAV's stringent power and hardware constraints.

To meet the high uplink throughput demands of LAE services, multi-carrier MIMO transmission is often required, necessitating reliable channel estimation (CE) and signal equalization (SE). Conventionally, the receiver performs CE and SE in a sequential (decoupled) manner, i.e., the channel is first estimated from pilots and then used for equalization [3–7].

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However, in the considered UAV-assisted multi-carrier MIMO system, this approach incurs prohibitive fronthaul and computational overhead due to limited UAV-to-GBU bandwidth [8, 9]. To address this, we employ a linear compression (LC) scheme for joint CE and SE in a reduced-dimensional domain. Since LC applies to both pilots and data, a compression-aware joint design is necessary. While LC schemes have been explored in wireless sensor networks, most existing studies focus on single-carrier settings [10–14]. Although [15] investigated an LC-based multi-carrier SE problem, it assumed perfect channel state information (CSI) and ignored practical pilot-based channel estimation.

To address the fronthaul and complexity issues in the UAV-assisted air-to-ground system, we investigate a linear compression-based uplink joint channel estimation and equalization (LC-CE) design problem under a multi-carrier MIMO setting in this paper. To reduce feedback and signaling overhead, the GBU computes a single shared compression matrix from the received pilot signals and feeds it back to the UAV, which then compresses high-dimensional data signals across all subcarriers before fronthaul transmission. Subsequently, the GBU performs equalization based on the estimated compressed CSI. We formulate this as a compression-based multi-carrier joint channel estimation and equalization problem, which is nontrivial due to the coupling between the shared compression matrix and the subcarrier-wise equalization matrices. To tackle this challenge, we impose an orthogonality constraint and reformulate the problem on the Grassmann manifold, which enables an efficient Riemannian gradient descent (RGD) algorithm. A low-complexity heuristic scheme is also developed for real-time implementation. Simulation results show that the proposed LC-CE scheme substantially reduces fronthaul overhead while approaching the performance of the uncompressed baseline.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

Consider an uplink UAV-assisted multi-carrier MIMO system, where the UAV is equipped with M antennas and serves L single-antenna users over N_{sc} subcarriers. All L UEs are simultaneously served by the UAV on the same time-frequency resource. We assume that an OFDM frame consisting of a pilot block and a data block is transmitted, during which

the corresponding channel coefficients are assumed to remain constant. The pilot block and the data block consist of L and N_d OFDM symbols, respectively. For the pilot block, the pilot sequences transmitted by different UEs are orthogonal.

1) *Channel Estimation*: For uplink channel estimation, the received pilot block at the j -th subcarrier is denoted as

$$\mathbf{Y}_j^p = \mathbf{H}_j \mathbf{X}_j^p + \mathbf{N}_j^p, \quad \forall j \quad (1)$$

where $\mathbf{H}_j \in \mathbb{C}^{M \times L}$ is the channel matrix between the users and the UAV, $\mathbf{X}_j^p \in \mathbb{C}^{L \times L}$ is the power-normalized pilot matrix with the ℓ -th column denoting UE ℓ 's pilot signal, and $\mathbf{N}_j^p$ is the additive white Gaussian noise (AWGN) matrix. By the MMSE criterion, the channel at the j -th subcarrier is estimated by solving

$$\min_{\mathbf{G}_j} \mathbb{E} [\|\mathbf{H}_j - \mathbf{G}_j \mathbf{Y}_j^p (\mathbf{X}_j^p)^H\|^2], \quad (2)$$

where $\mathbb{E}(\cdot)$ denotes the statistical expectation and the MMSE channel estimator $\mathbf{G}_j \in \mathbb{C}^{M \times M}$ is

$$\mathbf{G}_j = \mathbf{R}_{\mathbf{H}_j} (\mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{I}_M)^{-1} \quad (3)$$

where $\mathbf{R}_{\mathbf{H}_j} = \mathbb{E}\{\mathbf{H}_j \mathbf{H}_j^H\}$ is the channel covariance matrix, and the estimated channel is given by $\hat{\mathbf{H}}_j = \mathbf{G}_j \mathbf{Y}_j^p (\mathbf{X}_j^p)^H$.

2) *Uplink Equalization*: In the uplink transmission, the received data signal $\mathbf{y}_j \in \mathbb{C}^M$ on the j -th subcarrier is

$$\mathbf{y}_j = \hat{\mathbf{H}}_j \mathbf{x}_j + \Delta \mathbf{H}_j \mathbf{x}_j + \mathbf{n}_j, \quad \forall j \quad (4)$$

where $\Delta \mathbf{H}_j = \mathbf{H}_j - \hat{\mathbf{H}}_j \in \mathbb{C}^{M \times L}$ is the channel estimation error, $\mathbf{x}_j \in \mathbb{C}^L$ is the transmitted data symbol by users, and $\mathbf{n}_j \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_M), \forall j$, is the AWGN noise matrix.

Conventionally, the equalization problem is to estimate the transmitted signals on each subcarrier by solving the following MSE minimization problem

$$\min_{\{\mathbf{W}_j\}} \mathbb{E} [\|\mathbf{x}_j - \mathbf{W}_j \mathbf{y}_j\|_2^2 | \hat{\mathbf{H}}_j], \quad \forall j \in \mathcal{N} \quad (5)$$

where $\mathbf{W}_j \in \mathbb{C}^{L \times M}$ is the linear equalization matrix on subcarrier j . The optimal solution to problem 5 is obtained by taking the first-order derivative of the objective function and utilizing the property $\mathbb{E}[\hat{\mathbf{H}}_j \Delta \mathbf{H}_j^H] = \mathbf{0}$, which is given by

$$\mathbf{W}_j = \hat{\mathbf{H}}_j^H (\hat{\mathbf{H}}_j \hat{\mathbf{H}}_j^H + \mathbf{C}_j + \sigma^2 \mathbf{I}_M)^{-1} \quad (6)$$

where $\mathbf{C}_j \triangleq \mathbb{E}[\Delta \mathbf{H}_j \Delta \mathbf{H}_j^H]$ is the covariance matrix of $\Delta \mathbf{H}_j$

$$\mathbf{C}_j = \mathbf{R}_{\mathbf{H}_j} - \mathbf{R}_{\mathbf{H}_j} (\mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{I}_M)^{-1} \mathbf{R}_{\mathbf{H}_j} \quad (7)$$

However, the conventional per-subcarrier processing in multi-carrier MIMO incurs excessive fronthaul and computational costs, since it requires fronthauling all pilots/data and performing signal processing with M -dimensional matrix operations on each subcarrier. This motivates practical low-complexity channel estimation and equalization schemes.

B. Linear Compression-based Scheme

In the UAV-assisted MIMO setting, the number of antennas M is typically much larger than the number of users L , revealing substantial spatial redundancy. We therefore consider compressing the received signal to a lower dimension and performing channel estimation and equalization in the compressed domain, which reduces both fronthaul overhead and computational complexity. Accordingly, we adopt a linear compression-based estimation and equalization framework.

1) *Compressed Channel Estimation*: In the LC-CE scheme, we first fronthaul the received pilot signal $\mathbf{Y}_j^p$ to the GBU, and then compress it to a lower dimension via a compression matrix for low-complexity channel estimation. Specifically, the compressed received pilot block is

$$\mathcal{Y}_j^p \triangleq \mathbf{V} \mathbf{Y}_j^p = \mathbf{V} \mathbf{H}_j \mathbf{X}_j^p + \mathbf{V} \mathbf{N}_j^p, \quad \forall j \quad (8)$$

Here, the compressor $\mathbf{V} \in \mathbb{C}^{r \times M}$ is shared by all subcarriers, where the compression dimension r is predefined and satisfies $r \leq M$. Assuming that $\mathbf{V}$ is given, the GBU subsequently performs channel estimation for the compressed channel $\mathcal{H}_j = \mathbf{V} \mathbf{H}_j \in \mathbb{C}^{r \times L}$ by solving

$$\min_{\mathcal{G}_j} \mathbb{E} [\|\mathcal{H}_j - \mathcal{G}_j \mathcal{Y}_j^p (\mathbf{X}_j^p)^H\|^2], \quad (9)$$

where the involved matrix operation dimension is reduced from $M \times L$ to $r \times L$. Similarly, the MMSE channel estimator for the compressed channel with a given $\mathbf{V}$ is

$$\mathcal{G}_j = \mathcal{R}_{\mathcal{H}_j} (\mathcal{R}_{\mathcal{H}_j} + \sigma^2 \mathbf{V} \mathbf{V}^H)^{-1} \quad (10)$$

where $\mathcal{R}_{\mathcal{H}_j} = \mathbf{V} \mathbf{R}_{\mathbf{H}_j} \mathbf{V}^H \in \mathbb{C}^{r \times r}$ is the covariance matrix for $\mathcal{H}_j$, and the estimated compressed channel is $\hat{\mathcal{H}}_j = \mathcal{G}_j \mathcal{Y}_j^p (\mathbf{X}_j^p)^H = \mathcal{G}_j \mathcal{Z}_j$.

2) *Compressed Uplink Equalization*: The compression-based equalization also has two steps. The received data signal $\mathbf{y}_j$ is first compressed to a lower dimension at the UAV as

$$\mathbf{y}_j \triangleq \mathbf{V} \mathbf{y}_j = \hat{\mathcal{H}}_j \mathbf{x}_j + \Delta \mathcal{H}_j \mathbf{x}_j + \mathbf{n}_j, \quad \forall j, \quad (11)$$

where $\mathbf{y}_j \in \mathbb{C}^r$ is the compressed received signal, $\Delta \mathcal{H}_j = \mathbf{V} \mathbf{H}_j - \hat{\mathcal{H}}_j$ is the compressed-channel estimation error, and $\mathbf{n}_j = \mathbf{V} \mathbf{n}_j$ is the compressed noise. Then, the equalizers, i.e., $\mathbf{U}_j$, for each subcarrier are designed separately based on $\hat{\mathcal{H}}_j$ and estimate the users' signal from the compressed signal $\mathbf{y}_j$. Finally, the joint compression and equalization problem is

$$\min_{\mathbf{V}} \left\{ \min_{\{\mathbf{U}_j\}} \sum_{j=1}^{N_{sc}} \mathbb{E} [\|\mathbf{x}_j - \mathbf{U}_j \mathbf{V} \mathbf{y}_j\|_2^2 | \hat{\mathcal{H}}_j] \right\} \quad (12)$$

and the estimated uplink signal is $\hat{\mathbf{x}}_j = \mathbf{U}_j \mathbf{V} \mathbf{y}_j, \forall j$.

The reason for reusing the same compressor $\mathbf{V}$ among all subcarriers is to reduce fronthaul overhead. Since $\mathbf{V}$ is computed at the GBU and must be delivered to the UAV, using a common $\mathbf{V}$ reduces the fronthaul cost of data transmission from $N_d N_{sc} M r$ to $N_d M r$. Compared with problem (5), problem (12) introduces an additional compression matrix for fronthaul reduction. Because the compressor is shared across subcarriers, the equalizers become mutually coupled, which makes the joint design challenging.

III. PROPOSED ALGORITHM

A. Problem Reformulation on Grassmann Manifold

In fact, problem (12) can be reduced to a single-variable optimization problem, which is summarized in Lemma 1.

Lemma 1 (*Variable Elimination*) Problem (12) can be reformulated as a single-variable optimization problem w.r.t. the compression matrix $\mathbf{V}$ as follows

$$\min_{\mathbf{V}} J_1(\mathbf{V}) \triangleq \sum_{j=1}^{N_{sc}} \text{Tr}[(\mathbf{I}_r + \hat{\mathbf{H}}_j^H(\mathbf{C}_j + \sigma^2 \mathbf{V}\mathbf{V}^H)^{-1} \hat{\mathbf{H}}_j)^{-1}] \quad (13)$$

where $\mathbf{C}_j \triangleq \mathbb{E}[\Delta \mathbf{H}_j \Delta \mathbf{H}_j^H]$ represents the estimation error covariance matrix for the compressed channel, written as

$$\mathbf{C}_j = \mathbf{R}_{\mathbf{H}_j} - \mathbf{R}_{\mathbf{H}_j}(\mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{V}\mathbf{V}^H)^{-1} \mathbf{R}_{\mathbf{H}_j}. \quad (14)$$

proof: For any fixed compression matrix $\mathbf{V}$, the inner minimization with respect to each $\mathbf{U}_j$ in (12) is a convex quadratic problem, which has a closed-form solution:

$$\mathbf{U}_j = \hat{\mathbf{H}}_j^H(\hat{\mathbf{H}}_j \hat{\mathbf{H}}_j^H + \mathbf{C}_j + \sigma^2 \mathbf{V}\mathbf{V}^H)^{-1}. \quad (15)$$

Substituting this back into (12) yields the objective $J_1(\mathbf{V})$, which completes the proof.

While the problem is simplified to a single-variable optimization form, minimizing $J_1(\mathbf{V})$ is still challenging. To efficiently solve it, we leverage certain properties of $J_1(\mathbf{V})$ and reformulate it into a tractable form.

Proposition 1 The objective function $J_1(\mathbf{V})$ in problem (13) is left-invariant, i.e., for an arbitrary invertible matrix $\mathbf{B}$,

$$J_1(\mathbf{V}) = J_1(\mathbf{B}\mathbf{V}), \quad \forall \mathbf{B} \in \mathbb{C}^{r \times r}, \det(\mathbf{B}) \neq 0. \quad (16)$$

Proposition 1 follows by substituting $\mathbf{B}$ into the objective function. This naturally implies that $J_1(\mathbf{V})$ depends solely on the row-space of $\mathbf{V}$: $\mathcal{V}_{\text{row}} \in \text{Gr}_{r,M}$, where $\text{Gr}_{r,M} := \{\mathcal{S} \subset \mathbb{C}^M : \dim(\mathcal{S}) = r\}$ is the Grassmann manifold of r -dimensional subspaces of $\mathbb{C}^M$. Therefore, without loss of generality, we can define the row orthogonal representation of $\mathbf{V}$ as $\tilde{\mathbf{V}} \triangleq (\mathbf{V}\mathbf{V}^H)^{-1/2} \mathbf{V}$ which satisfies $\tilde{\mathbf{V}}\tilde{\mathbf{V}}^H = \mathbf{I}_r$ and $J_1(\tilde{\mathbf{V}}) = J_1(\mathbf{V})$. This inspires us to introduce an orthogonal constraint to (13) and use the orthogonality to simplify the complex objective function $J_1(\mathbf{V})$. Specifically, we have

$$\mathbf{C}_j + \sigma^2 \mathbf{V}\mathbf{V}^H \big|_{\mathbf{V}\mathbf{V}^H = \mathbf{I}_r} \quad (17a)$$

$$= \mathbf{R}_{\mathbf{H}_j} - \mathbf{R}_{\mathbf{H}_j}(\mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{I}_r)^{-1} \mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{I}_r \quad (17b)$$

$$= \mathbf{R}_{\mathbf{H}_j}(\mathbf{I} - (\mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{I})^{-1} \mathbf{R}_{\mathbf{H}_j}) + \sigma^2 \mathbf{I} \quad (17c)$$

$$= \sigma^2(\mathbf{R}_{\mathbf{H}_j}(\mathbf{R}_{\mathbf{H}_j} + \sigma^2 \mathbf{I}_r)^{-1} + \mathbf{I}_r) \quad (17d)$$

$$= \sigma^2(\mathbf{G}_j + \mathbf{I}_r). \quad (17e)$$

Recall that $\hat{\mathbf{H}}_j = \mathbf{G}_j \mathbf{Y}_j^H (\mathbf{X}_j^H)^H = \mathbf{G}_j \mathbf{Z}_j$, $J_1(\mathbf{V})$ admits the simplified expression $J_2(\mathbf{V})$ as

$$J_2(\mathbf{V}) \triangleq J_1(\mathbf{V}) \big|_{\mathbf{V}\mathbf{V}^H = \mathbf{I}_r} \quad (18a)$$

$$= \sum_{j=1}^{N_{sc}} \text{Tr}[(\mathbf{I}_r + \frac{1}{\sigma^2} \mathbf{Z}_j^H \mathbf{G}_j^H (\mathbf{G}_j + \mathbf{I}_r)^{-1} \mathbf{G}_j \mathbf{Z}_j)^{-1}] \quad (18b)$$

$$= \sum_{j=1}^{N_{sc}} \text{Tr}[(\mathbf{I}_r + \frac{1}{\sigma^2} \mathbf{Z}_j^H \mathbf{A}_j \mathbf{Z}_j)^{-1}]. \quad (18c)$$

Crucially, $J_2(\mathbf{V})$ remains invariant under left unitary transformations when $\mathbf{V}\mathbf{V}^H = \mathbf{I}_r$. This invariance confirms that $J_2(\mathbf{V})$ depends solely on the underlying row space, leading to the following proposition.

Proposition 2 Solving problem (19) is equivalent to solve problem (13).

$$\min_{\mathbf{V}: \mathbf{V}\mathbf{V}^H = \mathbf{I}_r} J_2(\mathbf{V}), \quad (19)$$

By Proposition 2, problem (19) can be reformulated with a more tractable objective. The remaining challenge is the nonconvex row-orthogonality constraint. Since the objective depends only on the row subspace, it is natural to optimize it over the Grassmann manifold, which removes rotational redundancy and enables efficient Riemannian methods like Riemannian gradient descent (RGD) described below.

B. Riemannian Gradient Descent

In this part, we derive all the Riemannian ingredients needed in manifold optimization for the compressor design problem on the Grassmann manifold.

1) *Orthogonal Projection:* From the perspective of manifold optimization, the Riemannian metrics of linear manifold $\mathcal{M} = \mathbb{C}^{r \times M}$ and Grassmann manifold $\widehat{\mathcal{M}} = \text{Gr}_{r,M}$ are

$$g_{\mathbf{V}}(\xi_{\mathbf{V}}, \zeta_{\mathbf{V}}) = \Re\{\text{Tr}(\xi_{\mathbf{V}}^H \zeta_{\mathbf{V}})\}, \quad (20a)$$

$$g_{\widehat{\mathcal{V}}_{\text{row}}}(\tilde{\xi}_{\mathbf{V}}, \tilde{\zeta}_{\mathbf{V}}) = \Re\{\text{Tr}(\tilde{\xi}_{\mathbf{V}}^H \tilde{\zeta}_{\mathbf{V}})\}, \quad (20b)$$

where $\xi_{\mathbf{V}}, \zeta_{\mathbf{V}}$ are tangent vectors in $\mathcal{M}$'s tangent space $T_{\mathbf{V}}\mathcal{M} = \mathbb{C}^{M \times r}$, and $\tilde{\xi}_{\mathbf{V}}, \tilde{\zeta}_{\mathbf{V}}$ are tangent vectors in $\widehat{\mathcal{M}}$'s tangent space $T_{\widehat{\mathcal{V}}_{\text{row}}}\widehat{\mathcal{M}} \triangleq \{\tilde{\xi} \in \mathbb{C}^{M \times r} : \mathbf{V}\tilde{\xi} = \mathbf{0}\}$. With the Riemannian metrics, both $\mathcal{M}$ and $\widehat{\mathcal{M}}$ are Riemannian manifolds. Let $\xi \in T_{\mathbf{V}}\mathcal{M}$, the orthogonal projection of ξ onto $T_{\widehat{\mathcal{V}}_{\text{row}}}\widehat{\mathcal{M}}$ is defined as [16]

$$\Pi_{T_{\widehat{\mathcal{V}}_{\text{row}}}\widehat{\mathcal{M}}}^{T_{\mathbf{V}}\mathcal{M}}(\xi) = (\mathbf{I}_M - \mathbf{V}^H \mathbf{V})\xi \quad (21)$$

2) *Riemannian Gradient:* The Riemannian gradient on the Grassmann manifold is obtained by orthogonally projecting the Euclidean gradient from $T_{\mathbf{V}}\mathcal{M}$ onto $T_{\widehat{\mathcal{V}}_{\text{row}}}\widehat{\mathcal{M}}$. Specifically, let $\nabla J_2(\mathbf{V})$ denote the Euclidean gradient of the objective function $J_2(\mathbf{V})$, which is given by

$$\nabla J_2(\mathbf{V}) = -\frac{2}{\sigma^2} (\mathbf{Y}_j \mathbf{X}_j^H \mathbf{P}_j \mathbf{Z}_j \mathbf{A}_j + \mathbf{R}_{\mathbf{H}_j} \mathbf{V}^H \mathbf{M}_j), \quad (22)$$

where

$$\mathbf{P}_j = (\mathbf{I}_r + \frac{1}{\sigma^2} \mathbf{Z}_j^H \mathbf{A}_j \mathbf{Z}_j)^{-2}, \quad (23a)$$

$$\mathbf{M}_j = \mathcal{L}_f(\mathbf{A}_j, \mathbf{Z}_j^H \mathbf{A}_j \mathbf{Z}_j). \quad (23b)$$

Here, $\mathcal{L}_f(\mathbf{A}, \mathbf{B})$ is an operator defined for $r \times r$ Hermitian positive semi-definite matrices $\mathbf{A}$ and $\mathbf{B}$. Let $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$ be the eigenvalue decomposition (EVD) of $\mathbf{A}$, where $\mathbf{U}$ is the eigen-matrix and $\mathbf{\Lambda} = [\lambda_1, \dots, \lambda_r]$ with $\lambda_1 \geq \dots \geq \lambda_r$, the operator is defined as

$$\mathcal{L}_f(\mathbf{A}, \mathbf{B}) = \mathbf{U}[\mathbf{F} \odot (\mathbf{U}^H \mathbf{B} \mathbf{U})] \mathbf{U}^H, \quad (24a)$$

$$\mathbf{F}_{i,j} = \begin{cases} \frac{f(\lambda_i) - f(\lambda_j)}{\lambda_i - \lambda_j} & i \neq j, \\ f'(\lambda_i) & i = j, \end{cases} \quad (24b)$$

$$f(x) = \frac{x^2}{(x + \sigma^2)(2x + \sigma^2)}. \quad (24c)$$

Then, the Riemannian gradient of $J_2(\mathbf{V})$ is given by

$$\text{grad} J_2(\mathbf{V}) = (\mathbf{I}_M - \mathbf{V}^H \mathbf{V}) \nabla J_2(\mathbf{V}). \quad (25)$$

3) *Retraction*: The operation of moving along the tangent vector while remaining on the manifold is formally defined by the retraction mapping. To maintain the row-orthogonality required by (19), we employ the QR-based retraction:

$$R_{\hat{\mathbf{V}}_{\text{row}}}(\boldsymbol{\xi}) = \text{qf}(\mathbf{V}^H + \boldsymbol{\xi})^H \quad (26)$$

where $\text{qf}(\cdot)$ denotes the Q-factor of the QR decomposition. This ensures that the updated $\mathbf{V}$ remains a row-orthogonal representative of the new subspace.

With the Riemannian ingredients derived above, we propose the following compressor design method using the RGD, referred to as LC-RGD. For the step-size $\alpha^{(t)}$ selection in RGD, we can obtain it through Armijo's backtracking procedure. More precisely, let $\mathbf{V}^{(t)}$ denote the t -th update, with the corresponding row space denoted by $\hat{\mathbf{V}}_{\text{row}}^{(t)}$, and let $\boldsymbol{\eta}^{(t)} = R_{\hat{\mathbf{V}}_{\text{row}}^{(t)}}(\alpha_0 \beta^n \boldsymbol{\xi}^{(t)})$ be the descent direction, we find the smallest integer $n \geq 0$ such that

$$J_2(\boldsymbol{\eta}^{(t)}) \geq J_2(\mathbf{V}^{(t)}) - c \alpha_0 \beta^n \|\boldsymbol{\xi}^{(t)}\|_2^2 \quad (27)$$

where $\alpha_0 > 0$ and $c, \beta \in (0, 1)$. Then, the step-size is given by $\alpha^{(t)} = \alpha_0 \beta^n$. As shown in [16], the LC-RGD is guaranteed to converge to a local optimum regardless of the initialization.

Algorithm 1 The proposed LC-RGD algorithm

- 1: Initialize $\mathbf{V}^{(0)}$; $\alpha_0 > 0$; $c, \beta \in (0, 1)$; $t = 0$.
 - 2: **repeat**
 - 3: Calculate the search direction $\text{grad} J_2(\mathbf{V}^{(t)})$ with (25)
 - 4: Set the search direction $\boldsymbol{\xi}^{(t)} = -\text{grad} J_2(\mathbf{V}^{(t)})$.
 - 5: Find step-size $\alpha^{(t)}$ via Armijo's backtracking rule in (27).
 - 6: Set $\boldsymbol{\eta}^{(t)} = R_{\hat{\mathbf{V}}_{\text{row}}^{(t)}}(\alpha^{(t)} \boldsymbol{\xi}^{(t)})$ and update $\mathbf{V}^{(t+1)} = \boldsymbol{\eta}^{(t)}$
 - 7: **until** Convergence
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IV. LOW-COMPLEXITY HEURISTIC ALGORITHM

While LC-RGD simplifies the problem on the Grassmann manifold, it still requires a high-complexity EVD in each iteration. To further reduce the computational overhead, we propose a heuristic low-complexity design for the compression matrix, referred to as LC-maxSNR. Specifically, we design the compressor by solving the following received SNR maximization problem for the compressed signal

$$\max_{\mathbf{V}: \mathbf{V}\mathbf{V}^H = \mathbf{I}_r} \frac{\sum_{j=1}^{N_{sc}} \text{Tr}(\mathbf{V} \mathbf{R}_{\mathbf{H}_j} \mathbf{V}^H)}{N_{sc} \sigma^2 \text{Tr}(\mathbf{V} \mathbf{V}^H)} \quad (28)$$

Here, the row-orthogonality constraint ensures that $\mathbf{V}$ is full row rank, and $\mathbf{V}$ is formed by the eigenvectors associated with the r largest eigenvalues of $\sum_{j=1}^{N_{sc}} \mathbf{R}_{\mathbf{H}_j}$. In contrast to the iterative nature of the LC-RGD algorithm, the LC-maxSNR algorithm requires only a single EVD, making it more suitable for computation sensitive scenarios.

V. SIMULATION

In this section, we evaluate the proposed LC-CE scheme. The fronthaul cost is measured by the ratio between the fronthaul overhead of the proposed LC-CE scheme and that of the conventional scheme without compression, i.e., $\frac{N_{sc} M L + M r + N_{sc} r N_d}{N_{sc} M (L + N_d)}$. According to [10], the LC-based scheme is performance lossless when $r \geq L$ in the single-carrier case, while it does not hold in our multi-carrier setting since the compressor is reused across subcarriers. Hence, we set $r = L$ in the following simulations. The other parameters are set as $M = 32$, $L = 4$, $N_{sc} = 24$ or 48 , $N_d = 10$, $\alpha_0 = 1$, $c = 10^{-4}$, and $\beta = 0.5$, with the maximum RGD iteration number set to 10. The pilot sequences are generated from DFT sequences, the transmitted data streams use 16-QAM modulation, and $\text{SNR} = 10 \log(E_s/N_0)$, where E_s and N_0 denote the symbol power and noise power, respectively. The simulations are conducted under the "3GPP_38.901_UMa_NLOS" channel model generated by QuaDRiGa [17].

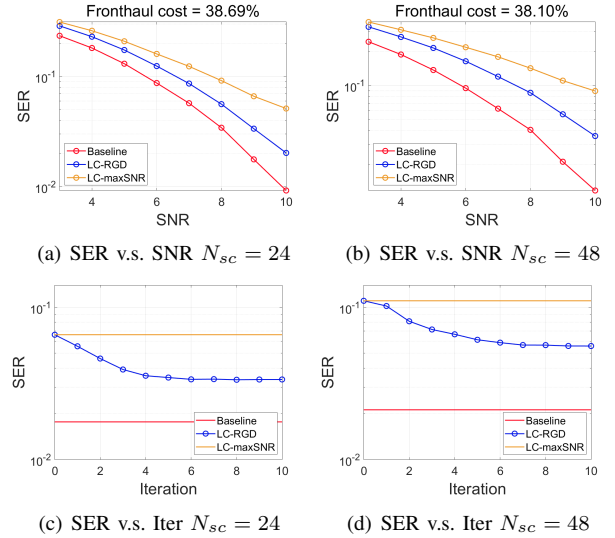


Fig. 1: SER performance of LC-CE scheme

Fig.1 compares the symbol error rate (SER) of the proposed LC-CE scheme with the uncompressed conventional baseline in Section II.A. For the LC-RGD algorithm, the LC-maxSNR scheme is used for initialization. As shown in Fig. 1(a), with $N_{sc} = 24$, LC-RGD approaches the baseline with a 1 dB loss at $\text{SER} = 10^{-1}$, while reducing the fronthaul cost to 39%. In Fig. 1(b), when $N_{sc} = 48$, the loss at $\text{SER} = 10^{-1}$ increases to 1.5 dB because more subcarriers share the same compressor. Fig. 1(c)-(d) further show that LC-RGD converges within 10 iterations.

VI. SUMMARY

In summary, we have investigated an LC-CE problem for UAV-assisted uplink with fronthaul-limited forwarding. We demonstrate that this problem can be formulated as a compressor design problem and solve it via two efficient methods, i.e., LC-RGD and heuristic LC-maxSNR. Simulations show large fronthaul reduction with performance close to the baseline.

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